

Loadscope — App Overview

A precision rifle load development tool

What it does

Loadscope is an app for precision rifle handloading. When shooters handload, there are many variables to consider — like how much powder to use, or how deep the bullet is seated into the brass. Each combination has to be tested at the range. After your range session, Loadscope reads the data a shooter already collects — a chronograph like the Garmin Xero for velocity, and your target groups, which you measure right in Loadscope from a photo or import from a target app such as BallisticX (more devices can be added). You import your chronograph data, the workbook fills in automatically, and then it does the math and tells the shooter which combination shot the best — so they have one more tool to decide for themselves which load they want to use.

Beyond a single test

Every load you crown a winner stays in a searchable Load Library, so a whole season of workups lives in one place — reopen any of them, print a Load Card for the bench, or start a fresh ladder off an old load without retyping it. Shot the same bullet over several powders? Compare Loads lines your saved loads up head-to-head — velocity SD, group, and vertical — and calls which one shot best. And a free iPhone and iPad companion, Loadscope Ballistics, carries your loads and a live DOPE dial to the firing line: measure a target group right there and send it back to the desktop over Wi-Fi.

How it scores and picks the best load

Loadscope uses a four-factor weighted scoring system to rank powder loads and seating depths. For powder ladder testing, Loadscope evaluates each load as the center of a three-charge "window" — plus the load just below it and the load just above. So Load 2 is scored inside the Load 1–Load 2–Load 3 group, Load 3 inside the Load 2–Load 3–Load 4 group, and so on, sliding one step at a time. The goal is to find a velocity flat-spot: a spot in the ladder where small changes in powder charge have little velocity change, which signals the rifle is insensitive to minor charge variations and will therefore shoot more consistently in the field. Four measurements are pulled for the middle charge of each window: (1) velocity spread across the three charges in fps (2) standard deviation of the shots fired at that charge in fps; (3) mean radius of the group on target, converted to MOA; and (4) vertical dispersion of the group, in MOA. Each measurement is then normalized on a 0-to-1 scale where 0 represents the best result among all candidates and 1 represents the worst — a min-max normalization across the full test set. The four normalized scores are multiplied by user-tunable weights — by default

30% velocity spread, 20% standard deviation, 20% mean radius, and 30% vertical — and summed into a single composite score. The lowest composite score wins, and that charge weight is displayed at the top of the Powder Charge Log as the suggested winner. Seating depth scoring follows the exact same pattern but ranks jump distances (S1 through S10) instead of charge weights, again surfacing the lowest-composite load as the recommended optimal seating depth.

How many formulas do this work

The Loadscope workbook contains 1,654 individual formula cells spread across five active sheets — the Powder Charge Log (powder ladder), the Charts sheet (where the scoring math actually runs), the Seating Depth Log, and the two import staging tabs (Garmin Xero Import and BallisticX Import). These 1,654 cells implement 37 distinct formula patterns; the same logical operations are repeated across rows so each load candidate gets evaluated identically. The patterns include LOOKUP-based routing of imported CSV data to the correct load row, IFERROR-wrapped min-max normalizations of the four scoring metrics, weighted-sum composite calculations, RANK-based winner identification, INDEX-MATCH pivots from rank back to the suggested charge weight, and a handful of MOA conversion calculations that translate linear inches at 100 yards into angular minute-of-angle measurements. Together they turn the raw chronograph and target-group numbers into a single, defensible recommendation that the shooter can take back to the range and confirm.